Run report: run-20261015T0728Z (D4 cyclotomic falsifier: exact yield + ecbench cost)

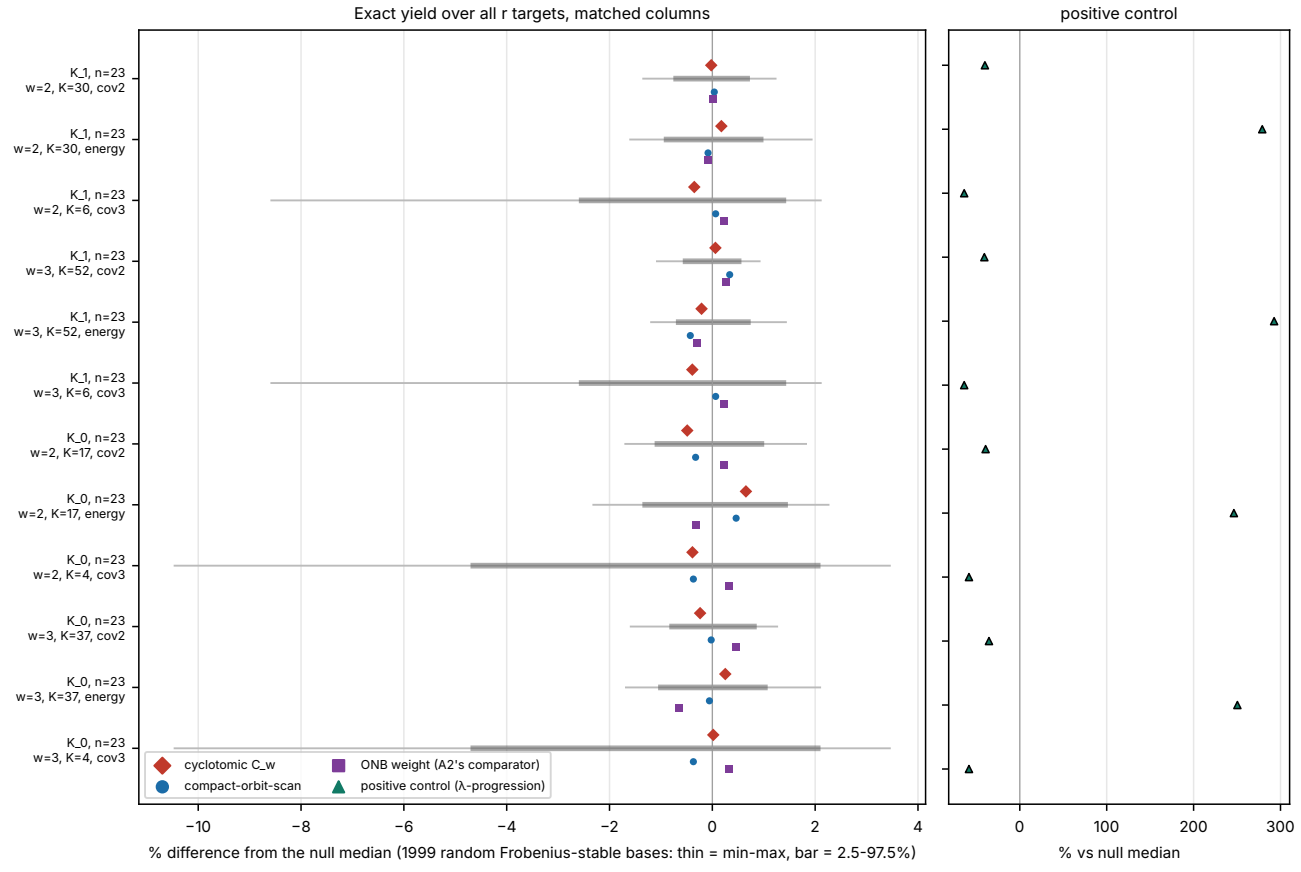
Commands. `research/cyclotomic_falsifier_20261015/run_yield.sh` (three invocations of `examples/cyclotomic_falsifier_yield.rs`, `--nulls 1999 --seed 20261015 --family 15`), then `run_cost.sh` (`ecbench run --wait --cpus none --allow-busy`, `verify --replay-all`, `table`, `compare --save`) on `specs/cost-a1-n23.json` and `specs/cost-a0-n23.json`. Protocol frozen at commit `9972ea0e8`; amendments `2984c8ca5`, `3512a53db`. Yield start 2026-10-15T07:28Z, cost start 07:36Z (UTC). Binaries: yield sha256 `be6c9ea3…311c`, ecbench `4d8bd56d…27f6`.

Host and isolation. 4 vCPU shared container, five lanes running; one thread, `nice 10`, 2.5 GB address-space cap; ecbench L0 on every record. Wall times descriptive only.

Workloads. Yield: $K_1/F_{2^{23}}$ (r = 4196903, h = 2), $K_0/F_{2^{23}}$ (r = 2095853, h = 4), $K_1/F_{2^{11}}$ (r = 991); exact over all r targets; 1999 null bases per rung. Cost: 16 public targets per curve (seeds 202610151001, 202610151003), 2 rounds + 1 warm-up.

Completion. Yield: 3/3 rungs written (349 s, 125 s, < 1 s); all self-checks passed. Cost: 240/240 records verified on each session; replay-all 160/160 identical on each; audits `ok` (receipts `62cb9d9c…cbf0`, `b7590fbe…0e4f`). The earlier session `sessions/void-a1-n23-full-unpriced` errored on every record (missing `full_prices.json` row) and was stopped (amendment A2).

Measured values (yield, n = 23). All 15 cyclotomic tests: effect vs null median between −0.49% and +0.65%, two-sided rank p between 0.363 and 1 (critical 0.00333). Positive control: p = 0.001 on all 12 n = 23 tests (−35% to −64% coverage, +246% to +293% energy). Full table: study README §3; raw: `yield/yield-*.json`.



Black edge = significant at p < 0.05/15 (two-sided rank). Source: yield/yield-*.json (examples/cyclotomic_falsifier_yield.rs, seed 20261015). K=1 cells omitted: identical for every base; n=11 in the README table.

Figure 1: Exact coverage and energy relative to the null median, n = 23 cells; thin line = min–max of 1999 null bases, bar = 2.5–97.5%; seed 20261015.

Measured values (cost, mean S, 95% interval as `ecbench table` recorded; IC values are lower bounds).

session	curve	arm	mean S	95%	÷ rho-frob
ECBS1h6e7055fb8555	K1 n23	rho-frob	0.439	[0.375, 0.512]	1
		ic-cyc-w1	4.562	[3.021, 6.754]	10.40
		ic-cos-c1	5.162	[3.825, 6.637]	11.76
		ic-cyc-w2	13.154	[13.146, 13.161]	29.98
		ic-cos-c30	12.872	[12.864, 12.880]	29.34
ECBS1h6c2295d00f35	K0 n23	rho-frob	0.385	[0.323, 0.456]	1
		ic-cyc-w2	7.042	[7.032, 7.054]	18.29
		ic-cos-c17	6.276	[6.264, 6.288]	16.30

		ic-cyc-w3-c3	1.028	[0.913, 1.147]	2.67
		ic-cos-c3	1.052	[0.937, 1.183]	2.73

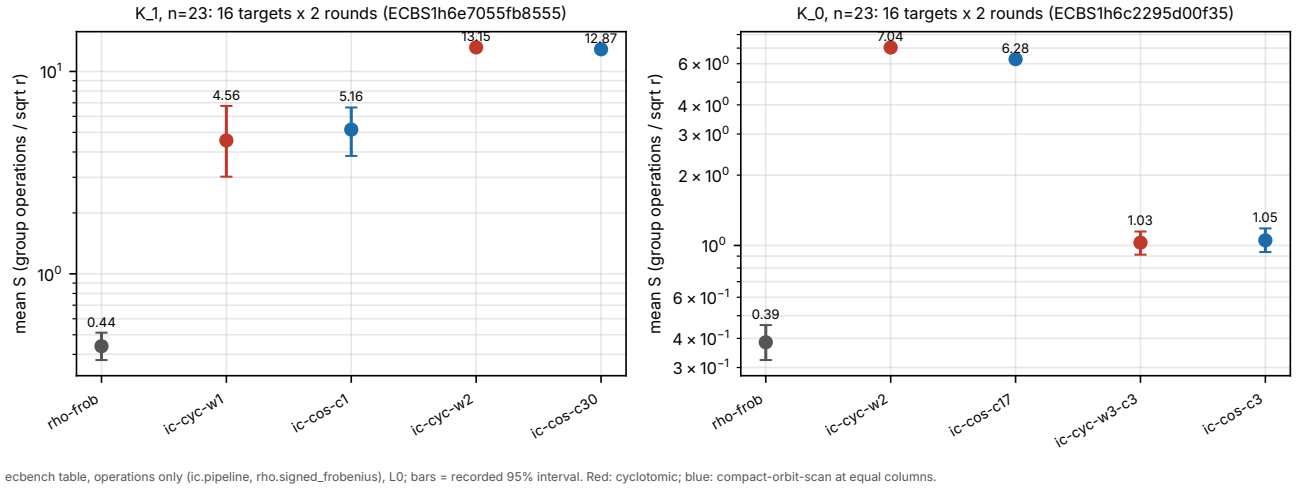


Figure 2: Mean S per arm, sessions as in the table; bars = recorded 95% interval.

Raw records. `session-archives/<curve>.tar.gz` (the full session directories; `records.jsonl` inside), `audits/audit-<curve>.json`, `tables/`. Interpretation is in the study README and REPORT.pdf.